

MAENG510 – TECHNICAL REPORT WRITING

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Future Stock Market Prediction Using Deep Reinforcement Learning

Abstract:

The stock market is a highly dynamic and complex environment influenced by numerous economic, political, and psychological factors, making accurate prediction a challenging task. Traditional machine learning models, while effective in capturing linear and nonlinear patterns, often fail to adapt to the continuously changing market conditions. This study explores the use of Deep Reinforcement Learning (DRL) for future stock market prediction. By modeling the prediction task as a sequential decision-making problem, the DRL agent learns optimal trading strategies through trial-and-error interactions with the environment. Historical stock data, including technical indicators such as moving averages and relative strength index (RSI), are used to train and evaluate the model. The proposed approach not only predicts future price trends but also adapts dynamically to market fluctuations. Experimental results demonstrate that DRL offers a robust and adaptive framework for financial forecasting, outperforming conventional regression-based methods in terms of predictive accuracy and adaptability.

Keywords: *Stock Market Prediction, Deep Reinforcement Learning, Financial Forecasting, Sequential Decision-Making, Machine Learning, Time Series Analysis, Adaptive Trading Strategy*